INTERNAL HOM COMMUTES WITH FLAT BASE CHANGE IN QCOH OF ALGEBRAIC STACKS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Pullback and internal hom on ringed sites. Stacks Project, Modules on Sites, Lemma modules-lemma-flat-pushforward-pullback-hom (internal hom pullback identity). Gives the sheaf identity $f^*\mathcal{H}om(\mathcal{G},\mathcal{H}) = \mathcal{H}om(f^*\mathcal{G}, f^*\mathcal{H})$.
- Flat morphisms and hom in QCoh^{fbc}. Stacks Project, Cohomology of Algebraic Stacks, Section on flat morphisms, Lemma stacks-cohomology-lemma-flat-morphism-and-Q. Identifies hom of the qc category with sheaf Hom under flat base change.

Source reference. Original lemma 0GQN: sources/stacks-chow/tags.txt, source lines 20868–20868. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0GN2: sources/stacks-chow/tags.txt, source lines 20777–20777. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06TX: sources/stacks-chow/tags.txt, source lines 8726–8726. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GQJ: sources/stacks-chow/tags.txt, source lines 20864–20864. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a flat morphism of algebraic stacks. Let $\mathcal{F}$ be an $\mathcal{O}_\mathcal{Y}$-module of finite presentation and let $\mathcal{G}$ be a quasi-coherent $\mathcal{O}_\mathcal{Y}$-module. Then $f^*hom(\mathcal{F}, \mathcal{G}) = hom(f^*\mathcal{F}, f^*\mathcal{G})$ with notation as in Lemma 0GQN.*

Proof. We have $f^* \mathcal{H}om_{\mathcal{O}_\mathcal{Y}}(\mathcal{F}, \mathcal{G}) = \mathcal{H}om_{\mathcal{O}_\mathcal{X}}(f^*\mathcal{F}, f^*\mathcal{G})$ by Modules on Sites, Lemma 0GN2. (Observe that this step is not where the flatness of f is used as the morphism of ringed topoi associated to f is always flat, see Sheaves on Stacks, Remark 06TX.) Then apply Lemma 0GQJ (and here we do use flatness of f). □

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